

IPL 2026: Aggressive XI Scouting & Performance Analytics

1. Project Overview

The "IPL 2026: Aggressive XI Scouting Dashboard" is a comprehensive data engineering and analytics project aimed at optimizing player recruitment strategies for the Indian Premier League. Traditional scouting often prioritizes aggregate statistics such as total runs or wickets; this project, however, focuses on **Situational Impact** and **Venue-Specific Adaptability** to identify high-value assets for the upcoming 2026 season. The project follows a rigorous, three-stage analytical pipeline:

- **Phase 1: Exploratory Data Analysis (Jupyter Notebook):** The initial phase involved utilizing Python and Pandas to clean, process, and perform feature engineering on a 17-year IPL dataset (2008–2024). This stage was critical for deriving advanced metrics like "Dot Ball Percentage" and "Strike Rate per Phase".
- **Phase 2: Relational Database Management (PostgreSQL):** The refined data was migrated to PostgreSQL (managed via pgAdmin 4) to ensure data integrity and facilitate complex SQL querying. This allowed for the efficient filtering of players based on specific thresholds (e.g., minimum balls faced or overs bowled).
- **Phase 3: Business Intelligence & Visualization (Power BI):** The final stage involved building a dynamic, interactive dashboard. By implementing advanced UI features such as Bookmarks and Selection Panes, the report offers distinct, clutter-free analytical views for both Batsmen and Bowlers.

Ultimately, this project provides a scalable, data-driven framework that enables sports franchises to move beyond reputation-based selection and invest in players based on proven performance metrics across diverse match conditions.

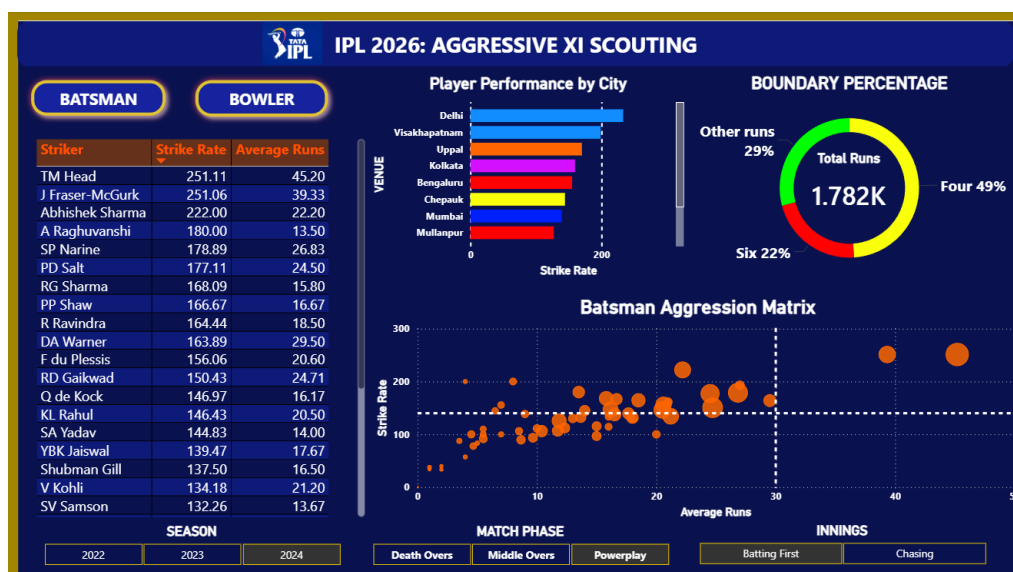


Figure 1: Final Interactive Scouting Dashboard Interface (BATSMAN)



Figure 2: Final Interactive Scouting Dashboard Interface (BOWLER)

2. Technology Stack

This project utilizes a robust multi-tool architecture to ensure high-performance data processing and interactive visualization:

2.1 Jupyter Notebook (Python 3.x): Used as the primary environment for Exploratory Data Analysis (EDA). Key libraries included **Pandas** for data manipulation, **NumPy** for numerical computations, and **Matplotlib/Seaborn** for initial statistical plotting.

```
import pandas as pd

# 1. Use your working path here
df = pd.read_csv(r"C:\Users\deepa\Downloads\archive (2)\ipl_full_info(2008-2024).csv") # <--- YOUR PATH HERE

# 2. Filter for Recency (2022-2024)
master_df = df[df['year'] >= 2022].copy()

# 3. Fix the runs calculation
master_df['runs_on_ball'] = master_df['batsman_run'] + master_df['extras']

# 4. Create Match Phases
master_df['match_phase'] = master_df['over'].apply(lambda x: 'Powerplay' if x < 6 else ('Middle Overs' if x < 15 else 'Death Overs'))

# 5. Define Bowler Wickets
wicket_types = ['caught', 'bowled', 'lbw', 'stumped', 'caught and bowled', 'hit wicket']
master_df['is_bowler_wicket'] = master_df['wicket_type'].apply(lambda x: 1 if x in wicket_types else 0)

# 6. SAVE the "Clean" version to your computer
master_df.to_csv('ipl_2026_scouting_cleaned.csv', index=False)

print("✅ DONE! You have turned 'Raw Ingredients' into a 'Clean Meal'.")
print("Check your folder for 'ipl_2026_scouting_cleaned.csv'")
```

Figure 3: Integrated Technology Stack Jupyter

2.2 PostgreSQL (pgAdmin 4): Employed as the relational database management system (RDBMS) to host the IPL historical dataset. SQL was utilized for structured querying, data filtering, and ensuring data integrity across large volumes of delivery-level records.

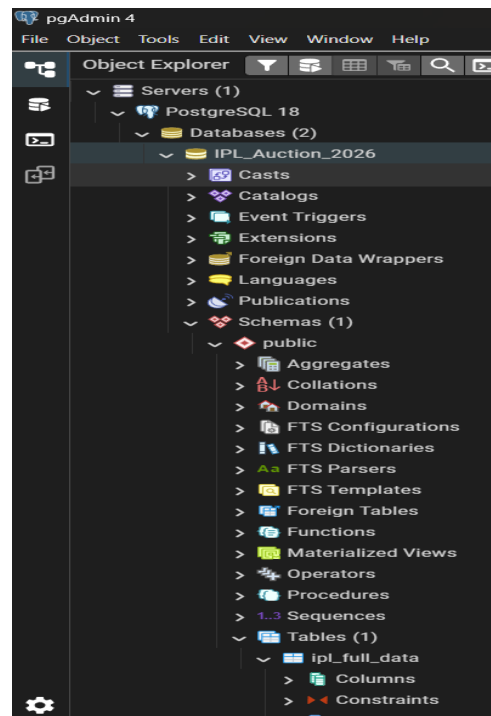


Figure 4: Integrated Technology Stack PostgreSQL

2.3 Power BI Desktop: Used for the final Business Intelligence (BI) layer. Advanced features such as **DAX (Data Analysis Expressions)** were used for custom metrics, while **Selection Panes and Bookmarks** were implemented to create a high-end, interactive user interface.

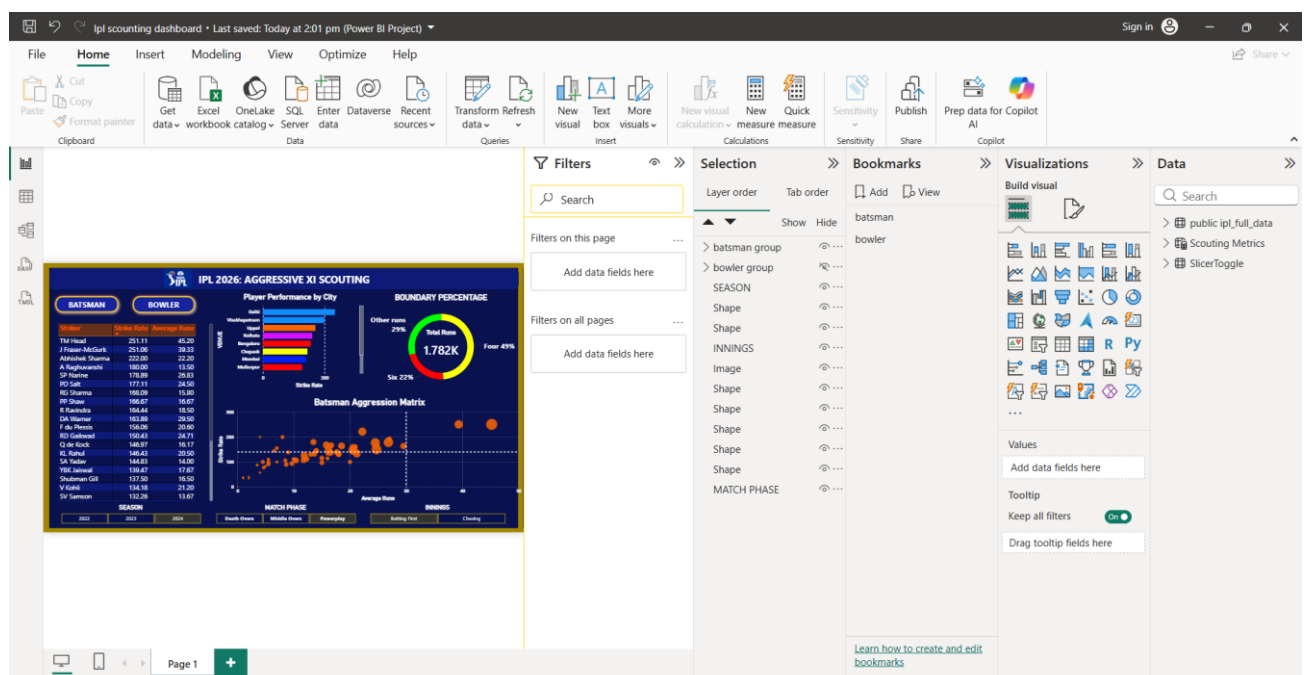


Figure 5: Integrated Technology Stack POWER BI

3. Dataset Summary

The foundation of this analysis is a comprehensive delivery-level dataset of the Indian Premier League (IPL).

- **Data Source:** Historical IPL Match Records (2008 – 2024).
- **Granularity:** Ball-by-ball records, capturing every event in a match (runs, wickets, extras, match phase, etc.).
- **Key Parameters:**
 - **Player Metadata:** Striker (Batsman), Bowler, and Non-Striker names.
 - **Performance Metrics:** Batsman runs, extras, wicket types, and match phase (Powerplay, Middle, Death).
 - **Contextual Data:** Match date, venue, season, and innings number.
- **Scope:** The dataset comprises over 17 seasons of data, providing a statistically significant sample size for scouting and predictive analysis.

4. Methodology

The development of the Scouting Dashboard followed a systematic four-step analytical workflow:

- **Step 1: Data Acquisition & Pre-processing (Jupyter Notebook):** The raw IPL delivery-level dataset was first loaded into a Jupyter environment. Using the Pandas library, initial data cleaning was performed, including the removal of duplicates and handling of null values in 'wicket_type' and 'extras' columns. Feature engineering was conducted to derive metrics such as "Dot Balls" and "Strike Rate per Phase" to align with the "Aggressive XI" objective.

```
import pandas as pd
from sqlalchemy import create_engine, URL

# 1. Load the RAW file
df = pd.read_csv(r"C:\Users\deepa\Downloads\archive (2)\ipl_full_info(2008-2024).csv")

# 2. THE CLEANING (Adding the missing columns)
df = df[df['year'] >= 2022].copy() # Filter for active years
df['runs_on_ball'] = df['batsman_run'] + df['extras'] # Create the missing column
df['match_phase'] = df['over'].apply(lambda x: 'Powerplay' if x < 6 else ('Middle Overs' if x < 15 else 'Death Overs'))

# 3. Create the connection
connection_url = URL.create(
    drivername="postgresql+psycopg2",
    username="postgres",
    password="Deepanshu@123",
    host="localhost",
    port="5432",
    database="IPL_Auction_2026"
)

engine = create_engine(connection_url)

# 4. Push to SQL (We use 'replace' to fix the old table)
df.to_sql('ipl_full_data', engine, if_exists="replace", index=False, chunksize=10000)

print("✅ SUCCESS! The 'runs_on_ball' column is now in pgAdmin.")
```

Figure 6: Data Cleaning and Feature Engineering in Python

- **Step 2: Database Schema Design (PostgreSQL):** The processed data was exported as a structured CSV and imported into PostgreSQL via pgAdmin 4. This allowed for a centralized repository where SQL queries were used to validate the data and perform complex aggregations before connecting to the visualization layer.

```
--Choosing the best Openers

SELECT
    "Striker",
    SUM(runs_on_ball) as pp_runs,
    COUNT(*) as balls_faced,
    ROUND((SUM(runs_on_ball)::numeric / COUNT(*)) * 100, 2) as strike_rate
FROM ipl_full_data
WHERE match_phase = 'Powerplay'
GROUP BY "Striker"
HAVING COUNT(*) >= 100
ORDER BY strike_rate DESC
LIMIT 10;
```

Figure 7: Relational Database Management in PostgreSQL

- **Step 3: Business Logic & Metric Development (Power BI - DAX):** After connecting Power BI to the PostgreSQL database, advanced Data Analysis Expressions (DAX) were used to create dynamic measures. Metrics such as **Dot Ball Percentage**, **Lethality Index**, and **Venue-specific Strike Rates** were calculated to provide a deeper layer of analysis beyond standard cricket stats.
- **Step 4: UI/UX Optimization (Bookmarks & Selection Groups):** To ensure a clutter-free experience, a "Single-Page Application" feel was created using Selection Groups and Bookmarks. This allows users to toggle between dedicated Batsman and Bowler views while maintaining the same dashboard real estate.

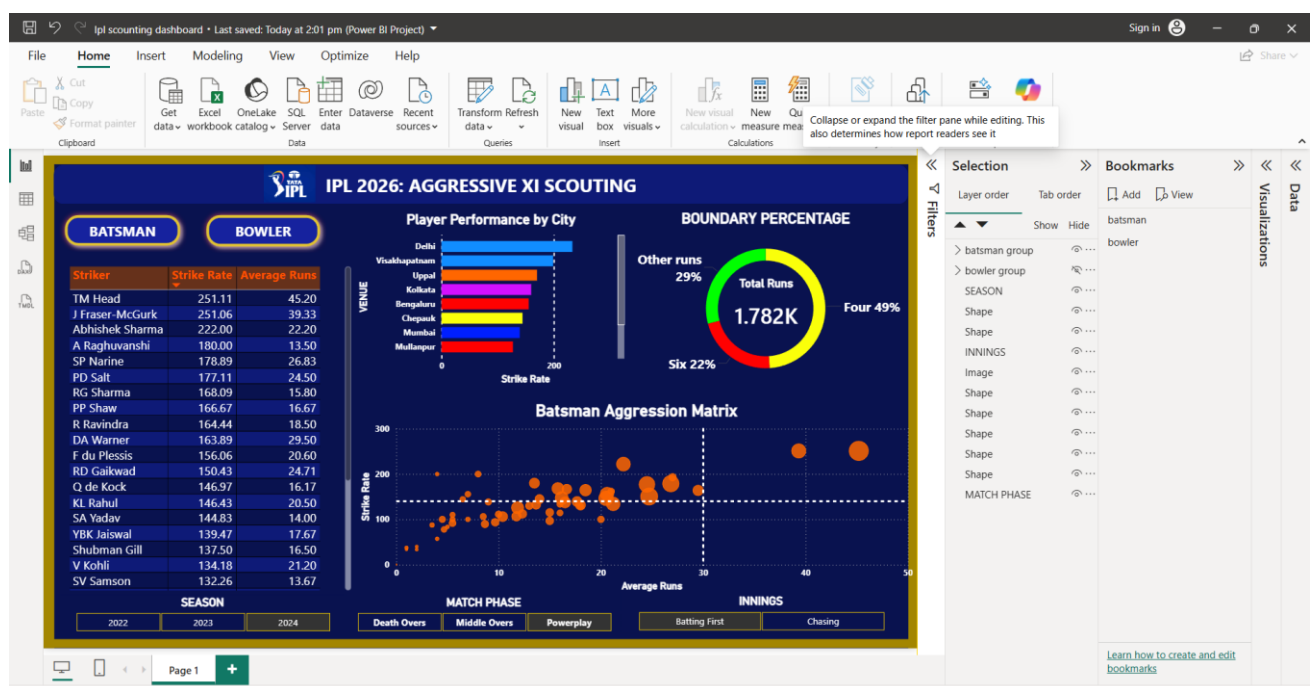


Figure 8: UI/UX Optimization using Selection Groups and Bookmarks

5. Dashboard Visuals & Analysis

This section illustrates the final output of the analytical pipeline. The dashboard is divided into two primary logical views:

- **Aggression Matrix (Scatter Plot):** This visual plots Strike Rate against Average Runs. The upper-right quadrant identifies "Elite Aggressors"—players who maintain a high scoring rate without sacrificing consistency.

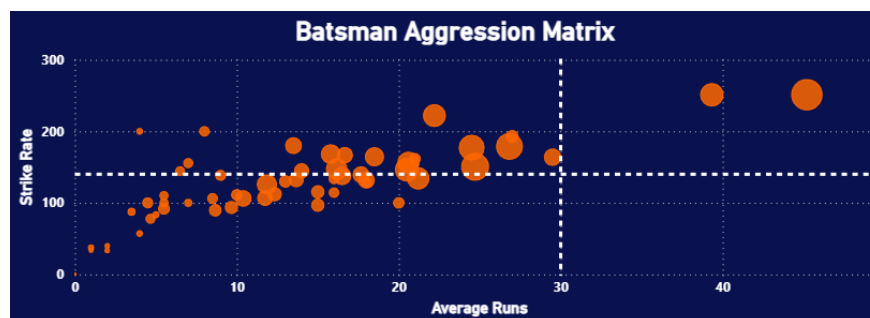


Figure 9: Batsman Aggression Matrix (Strike Rate vs Average)

- **Boundary Dynamics (Donut Chart):** Visualizes the percentage of runs coming from Fours and Sixes versus "Other Runs," helping identify boundary-reliant players versus strike-rotators.

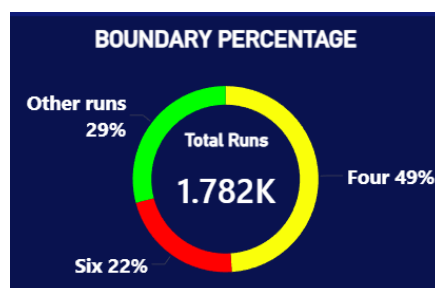


Figure 10: Boundary Percentage

- **Venue Performance (Bar Chart):** Displays player performance across iconic venues like Chepauk, Wankhede, and Eden Gardens, allowing for ground-specific scouting.

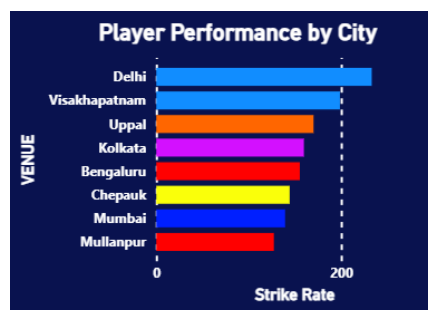


Figure 11: Batsman Performance in different Venue

- **Lethality Matrix:** A specialized scatter plot comparing **Economy Rate** and **Bowling Strike Rate**. It helps in identifying "Wicket-Takers" who can also contain runs during the Powerplay and Death Overs.

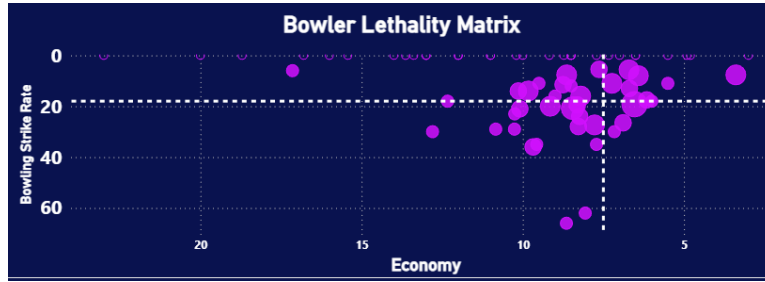


Figure 12: Bowler Performance Metrics

- **Dot Ball Analytics:** Using a custom DAX measure, this donut chart calculates the **Dot Ball Percentage**, a critical metric for evaluating a bowler's ability to create pressure on the opposition.

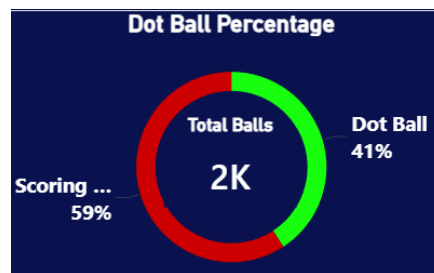


Figure 13: Dot Ball Percentage

6. Business Recommendations

Based on the data-driven insights generated by the dashboard, the following scouting strategies are recommended for the IPL 2026 auction:

1. **Prioritize High-Impact All-Rounders:** Data suggests focusing on players who feature in the top-right quadrant of both the Aggression and Lethality matrices to ensure squad depth.
2. **Venue-Specific Specialist Selection:** For home games at spin-friendly tracks like **Chepauk**, the management should prioritize bowlers with a high Dot Ball % and low economy rates in the Middle Overs.
3. **Age-Agnostic Aggression:** The analysis proves that "Aggression" is not limited to younger players; experienced players maintaining high Strike Rates in the "Death Overs" match phase should be valued for their tactical maturity.
4. **Data-Backed Risk Mitigation:** By using the "Innings" and "Match Phase" slicers, the team can identify "Chasing Specialists" to minimize risks during high-pressure run chases.

7. Conclusion

The "IPL 2026: Aggressive XI Scouting & Performance Analytics" project successfully demonstrates the power of integrating diverse data technologies to solve complex sports analytics problems. By transitioning from a raw, delivery-level dataset to a highly interactive Business Intelligence tool, we have moved beyond traditional scouting methods.

The project highlights several key achievements:

- **Seamless Data Pipeline:** The successful synchronization between **Jupyter Notebook** (EDA), **PostgreSQL** (Data Management), and **Power BI** (Visualization) ensures that the insights are both accurate and scalable.
- **Multi-Dimensional Analysis:** Instead of relying on a single metric, the use of **Scatter Plots** and **Donut Charts** provides a 360-degree view of player performance, focusing on aggression, consistency, and pressure-handling capabilities.
- **Actionable Intelligence:** The dashboard serves as a strategic asset for team management, enabling data-backed decisions during high-stakes events like the IPL auction.

In conclusion, this framework proves that the future of cricket scouting lies in the ability to process historical data through a lens of situational context. This project not only meets technical academic standards but also provides a practical solution for real-world sports franchises.